



3DEXPERIENCE®

Dymola 2015 FD01

Overview of new features

28 November 2014

 | The **3DEXPERIENCE®** Company

Executive Summary

Functional Mockup Interface

- FMI 2.0 specification finalized by Modelica Association.
- Full support for FMI 1.0 and FMI 2.0 in Dymola.

Modeling and simulation

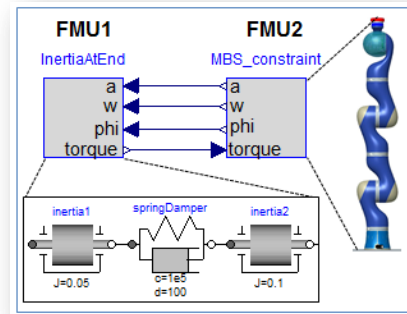
- Support for multi-core simulation speeds up certain kinds of models 2-3x.
- Convenient creation of “favorites” facilitates custom model packages.

Integration with external tools

- Can send commands to Dymola from Java, Python and JavaScript.
- Allows interactive web pages that run simulations and show results.

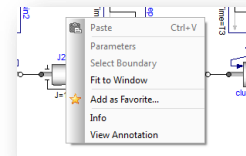
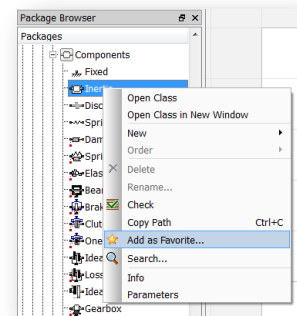
Functional Mockup Interface 2.0

- FMI 2.0 released in July 2014
 - Classification of interface variables
 - Save and restore FMI module state
 - Variable dependency information
 - Partial derivatives
 - Precise time event handling
 - Improved unit definitions
- Dymola 2015 FD01 fully compliant, both FMI 1.0 and 2.0



Creating “favorite” models

- Make your own collection to favorite models
 - Application-specific collections of models
 - Mix models from multiple packages
 - Can give additional parameter modifiers
- Use as any other Modelica package
 - Inserts reference to original model
 - Referenced packages opened when used
- New: easy creation of favorites
 - Right-click in the package browser or diagram layer



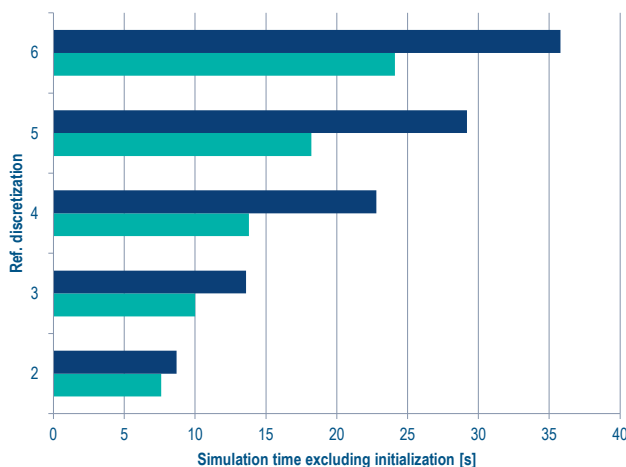
Multi-core simulation

- Parallelize the calculation of derivatives for continuous-time integration
`Advanced.ParallelizeCode = true`
- Requires compiler support for OpenMP
 - Visual Studio Pro 2010, Visual Studio Express 2012, or later
 - Certain other restrictions
- How well does it work?
 - Depends very much on model structure; uses all cores if possible
 - There is also an overhead because of multi-threading
 - Use log messages and block timers to check
- See also “Parallel Model Execution on many cores”, Modelica Conference 2014



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AirConditioning Evaporator example

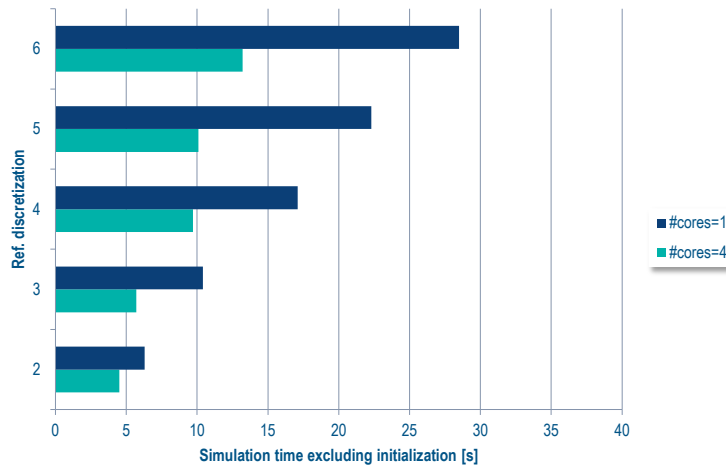


Intel Core i5 2520M
2 cores @ 2.5 GHz



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AirConditioning Evaporator example



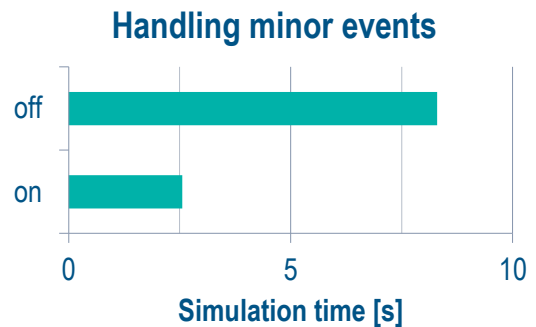
Intel Core i7 4800MQ
4 cores @ 2.7 GHz

More efficient event handling

- Turn on efficient handling of “minor” events
`Advanced.EfficientMinorEvents = true`
- Events that either do not influence the continuous model at all...
 - Sub-sampling signals in synchronous models
- ... or events that do not require a new Jacobian
 - Sampled input to a continuous model
- The efficient event handling is not approximate in itself, but...
 - Fewer Jacobian calculations
 - Slight changes in the simulation intervals for the numerical solvers
 - The results may change within the tolerance

Example of handling minor events

- David Harel's wrist watch from Modelica_StateGraph2 library
- Minor events have no impact on the continuous system
- Disable storing result at events (Simulation>Setup...>Output)



Intel Core i5 2520M @ 2.5 GHz

Tracing bus signals (expandable connector)

- Tracing of bus signals (expandable connectors) has been improved



- Name of signal clearly identified
- Shows where signal enters/exits expandable connector
- Viewing intermediate buses optional (by default only source and use)

JavaScript interface

- Call functions in the Dymola environment from JavaScript
 - Class DymolaInterface provides API for most common functions
 - General mechanism: `dymola.ExecuteCommand("a=1")`;

```
try {
  var dymola = new DymolaInterface();
  var result =
    dymola.simulateModel("Modelica.Mechanics.Rotational.Examples.CoupledClutches");
  if (result) {
    result = dymola.plot(["J1.w", "J2.w", "J3.w", "J4.w"]);
    if (result) {
      result = dymola.ExportPlotAsImage("C:/temp/plot.png");
    }
  } else {
    alert("Simulation failed.");
    var log = dymola.getLastErrorMessage();
    alert(log);
  }
} catch (e) {
  alert("Exception: " + e);
}
```

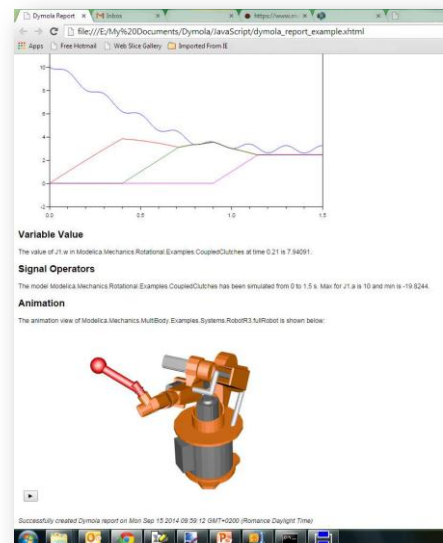
- Must start Dymola with command line argument `-serverport 8082`



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Report generator

- Interactive simulation functionality embedded in a webpage
 - Show model diagram
 - Change parameters
 - Simulate model
 - Show plots and animations
- Uses JavaScript and Dymola running as server



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System support

- Support for Windows 8.1
- FLEXnet upgraded to 11.11
 - Bug fixes and Windows 8 support
- Support for Visual Studio 2013 as C compiler
- Linux: 32- and 64-bit distributions



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Updated Modelica libraries

- | | | | |
|--------------------|-------|--------------------|-------|
| • Air Conditioning | 1.9 | • Hydro Power | 2.4.1 |
| • DataFiles | 1.0.2 | • Liquid Cooling | 1.3 |
| • Dymola Commands | 1.1 | • Model Management | 1.1.3 |
| • Electric Power | 2.2 | • Pneumatics | 1.6.3 |
| • Engine Dynamics | 1.2.2 | • Power Train | 2.3.0 |
| • Fuel Cell | 1.3 | • Thermal Power | 1.9 |
| • Heat Exchanger | 1.2 | • Vapor Cycle | 1.2 |
| • Hydraulics | 4.1 | • Vehicle Dynamics | 2.0 |



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